

★ Ovarro – Firmware Engineer (Official Job Description)

Role Overview

Ovarro is seeking a **Firmware Engineer** to work on technology for **critical infrastructure**.

You will help design, develop, and optimise embedded firmware for **next-generation IoT dataloggers and sensors** used in water, utilities, and industrial monitoring.

★ Key Responsibilities

- Design, develop, and maintain **embedded firmware** for **battery-powered IoT dataloggers**.
- Implement communication stacks for:
 - **LTE Cat 1**
 - **LTE-M / Cat-M1**
 - **NB-IoT**
 - 2G/3G fallback (where applicable)
- Integrate sensing technologies:
 - Pressure
 - Flow
 - Acoustic
 - Ultrasonic
 - Temperature
 - Analogue (0–10V, 4–20mA)
 - Digital pulse inputs
- Develop:
 - Efficient **data logging**
 - **Power-management** systems
 - **Wake/sleep** regimes
 - **OTA (over-the-air) update** mechanisms
- Deliver secure, reliable communication using:
 - **HTTPS**
 - **MQTT**
 - Similar secure protocols
- Work on **high-reliability**, field-deployed devices for utility networks.

Skills & Experience Required

- Strong experience in **embedded firmware development** for IoT devices.
- Proficiency in **C/C++**.
- Experience with microcontrollers:
 - **TI MSP**
 - **Atmel**
 - **STMicroelectronics (STM32)**
- Hands-on experience with **4G modems** (LTE Cat M1 / NB-IoT / Cat 1), including AT command stacks and vendor SDKs.
- Strong understanding of:
 - **Low-power design**
 - **Sensor integration**
 - **Analogue/digital interfaces**
 - **Wireless communication constraints**
- (Preferred) Knowledge of:
 - Clean water network operations
 - DMA monitoring
 - Leakage detection techniques

Location

- Wirksworth, Derbyshire (HQ)